

PHÂN TÍCH THỐNG KÊ NHIỀU CHIỀU (MAT3452)

Bài tập tuần 4: Vector ngẫu nhiên nhiều chiều, Mẫu ngẫu nhiên nhiều chiều

Trần Đăng Quang Minh - MSV: 23000141

Vector kì vọng: $\mu = (0.5, 1.0)^T$

Ma trận hiệp phương sai: $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

a. Sinh ngẫu nhiên 500 mẫu

```
library(MASS)
# Vector kì vọng
mu <- c(0.5, 1.0)
# Ma trận hiệp phương sai
sigma <- matrix(c(1, 0.5, 0.5, 1), nrow = 2)
n <- 500
set.seed(123)
samples <- mvrnorm(n = n, mu = mu, Sigma = sigma)
print(samples)
```

```
##           [,1]      [,2]
## [1,] 0.315560275 0.213667429
## [2,] 0.797509742 0.303811151
## [3,] 1.336488469 2.863273525
## [4,] 0.185531407 1.436592709
## [5,] 1.366549732 0.357383194
## [6,] 2.032863573 2.437716122
## [7,] 1.347139055 0.951191232
## [8,] 0.439800369 -1.130950702
## [9,] -0.169892084 0.480228047
## [10,] 0.153651267 0.574439558
## [11,] 1.608770567 2.011401299
## [12,] 0.703531644 1.419684186
## [13,] 0.405845675 1.788310840
## [14,] 0.493055292 1.198652796
## [15,] 0.326845378 0.210409535
## [16,] 2.414911797 2.180112545
## [17,] 0.997052558 1.365249765
## [18,] -1.358148910 -0.548131924
## [19,] 1.627232204 1.087551851
## [20,] 0.182705065 0.498396196
## [21,] -0.908396086 0.558871174
## [22,] 0.365368232 0.757088141
## [23,] -0.039335583 -0.237756250
## [24,] 0.006734263 0.230789095
## [25,] -0.598624157 1.016024388
## [26,] -1.235741236 -0.185697275
## [27,] 0.607206963 2.343882764
## [28,] 0.563276088 1.202373945
## [29,] -0.690793049 0.219482048
## [30,] 1.865064029 1.806607117
## [31,] 0.566643515 1.672014184
## [32,] 0.497627371 0.491293829
```

##	[33,]	1.985484314	1.064918810
##	[34,]	1.196489425	1.824482391
##	[35,]	0.238584479	2.684435697
##	[36,]	0.695922784	1.996837124
##	[37,]	0.397080065	2.062333455
##	[38,]	0.266955024	1.125810747
##	[39,]	0.539307150	0.430749971
##	[40,]	0.271622875	0.569382020
##	[41,]	0.034990161	0.261742055
##	[42,]	0.554288245	0.585588465
##	[43,]	-0.947949028	0.256218256
##	[44,]	2.977052717	2.279689214
##	[45,]	1.112942711	2.479308843
##	[46,]	-0.904716807	0.459435679
##	[47,]	0.750402677	0.051780318
##	[48,]	-0.223881390	0.915610608
##	[49,]	-0.039643720	2.890582933
##	[50,]	0.706408012	0.649192530
##	[51,]	0.296928148	1.641832389
##	[52,]	0.866378707	0.584176862
##	[53,]	-0.092482614	1.518228804
##	[54,]	1.560331986	2.310156706
##	[55,]	-0.521481105	1.630434287
##	[56,]	2.542787432	1.583816703
##	[57,]	-0.815610330	-0.366908216
##	[58,]	1.269752947	1.242827771
##	[59,]	0.705893356	1.008628487
##	[60,]	1.001800256	0.872221513
##	[61,]	1.245699227	0.911855646
##	[62,]	-0.224386059	0.854336316
##	[63,]	0.755224298	0.167643584
##	[64,]	-1.124127623	0.859903308
##	[65,]	0.164904863	-0.521301722
##	[66,]	0.712323939	1.313403090
##	[67,]	0.621666411	1.654655698
##	[68,]	0.252535337	1.339270676
##	[69,]	1.449580388	1.647833724
##	[70,]	2.235674420	2.815176415
##	[71,]	-0.405877540	1.055386612
##	[72,]	-0.771565949	-1.728031866
##	[73,]	1.761864968	1.480125256
##	[74,]	-0.274387034	0.546015280
##	[75,]	0.126558049	0.181776071
##	[76,]	0.703168863	2.573172857
##	[77,]	-0.083247590	1.090006273
##	[78,]	-0.593255926	-0.021089173
##	[79,]	1.410892078	0.403134760
##	[80,]	0.366666439	0.892766664
##	[81,]	0.663199866	0.846783997
##	[82,]	0.884835872	1.282489358
##	[83,]	0.769778610	0.088219383
##	[84,]	0.808717439	1.807375483
##	[85,]	0.828531256	0.289574816
##	[86,]	0.900442600	1.174220619
##	[87,]	1.259177535	2.140603364
##	[88,]	1.268636121	0.985120332

```

## [89,] -0.073760736 1.009230670
## [90,] 2.153151782 1.336641381
## [91,] 2.765286918 0.455512239
## [92,] 0.742441703 1.707409694
## [93,] 0.286477834 1.627017661
## [94,] 0.099140099 0.313294675
## [95,] 1.426296459 2.430422713
## [96,] 0.558118211 -0.097798314
## [97,] 2.457860242 2.830711635
## [98,] 2.798038924 1.356520548
## [99,] -0.294712944 1.386467947
## [100,] -1.318862005 1.041048856
## [101,] -0.652236261 0.921775999
## [102,] 0.736141301 1.208794335
## [103,] 0.303023737 0.769693395
## [104,] 0.957053092 -0.059014532
## [105,] -0.719318526 0.571066818
## [106,] 0.566371937 0.855637756
## [107,] 0.148624253 -0.008118673
## [108,] -0.238467193 -1.150492985
## [109,] 0.320595423 0.520832926
## [110,] 1.720404977 1.371343842
## [111,] 0.200250175 0.303219654
## [112,] 1.635312542 0.917712554
## [113,] -1.744922267 0.442667216
## [114,] 0.459883190 0.943880663
## [115,] 0.412347295 1.987292372
## [116,] 2.061656297 -0.040043373
## [117,] 0.818117184 0.864919354
## [118,] 0.282873466 0.107391175
## [119,] 0.375597540 -0.347328639
## [120,] -1.160226127 0.886383028
## [121,] 1.309525902 0.394243982
## [122,] -0.479732216 0.338658045
## [123,] -0.348053351 0.998382935
## [124,] 0.189122562 0.867312750
## [125,] 2.534459076 2.159203599
## [126,] -0.535188080 0.905977726
## [127,] 0.618556710 1.289144792
## [128,] 1.099265029 0.535767123
## [129,] 0.361032244 -0.527016804
## [130,] -0.605113331 1.981604103
## [131,] 2.090269313 1.911766168
## [132,] 1.818799807 0.463228153
## [133,] 0.269079077 1.302338439
## [134,] -0.021008118 0.789222138
## [135,] -0.601247086 -1.455081422
## [136,] 2.451244974 1.008288560
## [137,] -0.706800149 -0.323102666
## [138,] 0.571115196 2.210511488
## [139,] 1.835270172 2.971394207
## [140,] -0.503979446 -0.496916870
## [141,] 1.524857178 1.190668947
## [142,] 0.137396935 0.908463692
## [143,] -0.940193454 -0.282840107
## [144,] -1.126596540 0.003115207

```

```

## [145,] -0.689072036 -0.584869987
## [146,] -0.409455562  0.989898491
## [147,] -0.350511707 -0.681323234
## [148,]  1.261025751  1.430481051
## [149,]  1.948340431  3.189154955
## [150,] -1.109586892  0.380384717
## [151,]  2.151454204  0.712949503
## [152,]  1.112414914  1.719605321
## [153,]  0.483306367  1.592085378
## [154,]  0.352132397 -0.598691916
## [155,]  0.156238208  1.136863808
## [156,]  0.671257652  0.343083381
## [157,]  0.477436732  1.997689744
## [158,] -0.091782440  0.946699592
## [159,]  0.961557624  2.230609919
## [160,]  0.115243797  0.735963126
## [161,]  0.979850655  2.343499089
## [162,] -1.098871205  0.781643323
## [163,] -1.574450462  0.891797553
## [164,]  3.321020446  3.792625391
## [165,]  1.263516284 -0.485534806
## [166,]  0.742509668  1.274035673
## [167,]  0.948504905  1.654066113
## [168,]  0.158706362  0.503361015
## [169,]  0.663471350  1.731759971
## [170,]  0.314193672  1.824871636
## [171,]  0.572466224  0.554483794
## [172,]  0.703593089  0.909497762
## [173,]  0.271575788  1.169417997
## [174,]  2.618405286  2.568181545
## [175,] -0.187649584  0.403617800
## [176,]  0.531693190 -0.930014409
## [177,]  1.092675574  0.472775853
## [178,]  1.432761954  0.605006479
## [179,]  1.304852270  0.951228574
## [180,]  0.449696242  0.256391714
## [181,] -0.612020016  0.270285127
## [182,]  1.102893951  2.585006953
## [183,]  0.560885646  0.333502117
## [184,]  0.248863365 -0.247975618
## [185,]  0.816220320  0.274531461
## [186,]  0.536535030  0.621946303
## [187,]  1.580733701  1.841704633
## [188,]  0.331575881  1.315193415
## [189,]  1.313692155  1.492367313
## [190,]  1.106845064 -0.471644205
## [191,]  0.731432224  1.139997948
## [192,] -0.374779653  1.312407158
## [193,] -0.013888896  1.677712372
## [194,]  0.119074195 -0.169889023
## [195,]  0.138700845 -0.909075699
## [196,]  1.000607282  3.958667774
## [197,]  1.101440073  1.439018130
## [198,] -0.534907161 -0.132358411
## [199,] -0.239572304  0.681001885
## [200,]  0.280363859 -0.833675597

```

## [201,]	2.768335176	2.540116065
## [202,]	2.406804180	1.366361775
## [203,]	0.616924952	0.423830338
## [204,]	0.910995138	1.529844571
## [205,]	0.823525808	-0.041183650
## [206,]	-0.207433249	0.882549430
## [207,]	-0.327622105	0.461721923
## [208,]	0.437153854	0.032938828
## [209,]	1.816565335	2.542890277
## [210,]	0.079169690	1.327250852
## [211,]	0.072721777	1.633817031
## [212,]	0.817463644	1.104615365
## [213,]	1.614173817	2.020537023
## [214,]	0.096432680	0.509718545
## [215,]	-1.080267284	0.861194472
## [216,]	1.388660166	3.013732059
## [217,]	-0.299259337	1.035142231
## [218,]	0.017476901	0.230136101
## [219,]	-0.757264644	0.115976790
## [220,]	-0.814241618	0.089048713
## [221,]	0.523761033	-0.017912261
## [222,]	1.899343675	0.671039159
## [223,]	1.140241669	2.282071696
## [224,]	1.877444755	0.848134224
## [225,]	0.184221698	0.685905387
## [226,]	0.426621053	1.176868874
## [227,]	-0.392132132	0.671735258
## [228,]	-0.215842267	0.473583971
## [229,]	1.632556709	1.399702902
## [230,]	-0.872711903	0.613653957
## [231,]	1.324017363	3.562651129
## [232,]	-0.018808467	1.362370342
## [233,]	1.657621525	0.213970623
## [234,]	-0.839371846	1.060204339
## [235,]	0.030592776	0.474536830
## [236,]	-0.903026567	0.121887712
## [237,]	0.030565349	1.152598585
## [238,]	0.911192443	1.314506370
## [239,]	0.818487399	1.243224202
## [240,]	-0.686408986	0.832748083
## [241,]	-0.538767622	0.672834301
## [242,]	-0.430047971	1.060214275
## [243,]	0.604163198	3.487089893
## [244,]	-0.817141759	0.347274105
## [245,]	0.241246192	0.948627349
## [246,]	3.252810220	1.542177109
## [247,]	-0.933303817	2.258410186
## [248,]	-0.436318184	-0.418995006
## [249,]	-1.263074576	1.611660139
## [250,]	0.733098890	1.607742458
## [251,]	-0.594496728	1.443933471
## [252,]	0.068255956	0.458545635
## [253,]	-0.053576439	0.957894316
## [254,]	0.471393406	1.185351385
## [255,]	1.977409554	2.291288854
## [256,]	0.483497275	0.863103451

```

## [257,] 0.929582652 2.442416988
## [258,] 1.146978161 1.445520014
## [259,] 1.420426211 -0.117256284
## [260,] -0.729587452 -0.425476700
## [261,] -0.221196138 0.818594467
## [262,] -0.232468115 0.883987601
## [263,] 0.232553028 1.349120845
## [264,] 2.472055845 1.279954324
## [265,] 2.301493615 3.170235673
## [266,] 1.356314907 2.824174116
## [267,] -0.253601458 1.522977223
## [268,] -0.908716711 -0.633677983
## [269,] 0.324253054 0.502360468
## [270,] -0.166663195 1.821174637
## [271,] 2.065766751 0.897838705
## [272,] 1.551988661 1.615158684
## [273,] 0.863898310 1.821360390
## [274,] 0.100543851 -1.017229915
## [275,] 1.095998530 1.875626393
## [276,] -0.825661904 1.552202117
## [277,] 0.653413906 1.149353252
## [278,] 0.703790226 0.925336201
## [279,] 0.633347439 1.608259153
## [280,] 0.660905247 0.881833077
## [281,] -1.350775982 -0.037375608
## [282,] 0.685606483 2.090041948
## [283,] 0.832962984 1.335654645
## [284,] 0.858285023 0.181592865
## [285,] 1.261426512 0.443205784
## [286,] 0.912579555 0.819582189
## [287,] 0.292718224 1.590098725
## [288,] 2.900117051 1.441911876
## [289,] 1.253459387 -0.132865772
## [290,] 0.972438804 0.818658980
## [291,] 1.314652691 2.209047539
## [292,] 1.869730571 1.456164522
## [293,] 1.048452429 2.435201466
## [294,] -0.166786944 0.666583026
## [295,] 2.319520724 2.648881106
## [296,] 0.148350580 1.467178717
## [297,] 1.922558541 2.810923704
## [298,] -0.446948531 -0.392883558
## [299,] 0.402614852 1.133729786
## [300,] 1.258701092 2.406214450
## [301,] -0.297559576 0.558723768
## [302,] 0.177157336 0.019147128
## [303,] -0.740399462 0.614802743
## [304,] -0.987971351 0.664964875
## [305,] -0.016728540 0.759546017
## [306,] 0.714757246 1.358861908
## [307,] -1.206544919 -0.782170000
## [308,] -0.397127483 2.264288364
## [309,] 1.432834240 2.209149772
## [310,] 2.343737876 2.685443848
## [311,] 2.880810365 0.872892563
## [312,] 1.938027053 0.872745288

```

```

## [313,] -0.956555791 -0.534228991
## [314,] -0.124067112  0.582226933
## [315,]  0.056682596  0.833555055
## [316,]  0.698516182  2.020022962
## [317,]  0.505562145  0.811409736
## [318,] -1.197316084  0.517272711
## [319,]  2.419522136  1.998006093
## [320,]  1.893509372  1.185066651
## [321,]  1.320113736  0.591127559
## [322,]  1.183764489  2.426061513
## [323,] -0.617952572 -0.200872514
## [324,]  0.677378204  1.967196126
## [325,]  0.180997809  0.413291387
## [326,]  1.388087043  1.296194941
## [327,]  0.631502930  0.763150355
## [328,]  1.974468469  0.621851645
## [329,]  2.241399812  1.571784552
## [330,]  1.227330727  0.285296080
## [331,]  0.854070493  2.408392765
## [332,] -0.230549019 -0.327879111
## [333,] -0.519657703  0.769802149
## [334,]  1.057435831  3.073926433
## [335,]  0.922714976  1.230940167
## [336,] -1.419216550 -0.635337644
## [337,]  0.194242674 -1.056824845
## [338,]  0.735453429  0.416783651
## [339,]  1.221679534  1.777894383
## [340,]  0.262223061  1.061309964
## [341,]  1.420261266  1.160863149
## [342,] -0.011906480  3.172952519
## [343,]  2.176371002  2.217980863
## [344,]  0.516390136  1.080633692
## [345,]  0.130086415  1.279878282
## [346,] -1.005338774 -0.531357409
## [347,]  0.907803913  0.764236526
## [348,] -0.517889509  1.027416155
## [349,] -1.151289704  0.964255619
## [350,]  0.359043619  0.829349648
## [351,]  1.097832898  2.160100244
## [352,] -1.177064565 -0.774477064
## [353,] -0.378262326  1.138192892
## [354,]  1.179094548  0.522927153
## [355,] -1.433970559  1.386889665
## [356,]  1.090934054  0.987402807
## [357,]  1.585733470  0.626884056
## [358,]  0.646848760  0.795930933
## [359,] -1.708884721 -1.062176237
## [360,]  1.915137475  4.038758683
## [361,] -0.133399211  1.277810466
## [362,]  0.992720709  1.635179141
## [363,]  1.431830498  0.542346974
## [364,]  1.820411924  1.454374180
## [365,]  1.289756318  1.626471386
## [366,] -0.958199273  2.094826841
## [367,]  1.757616685  0.397389111
## [368,] -0.884275391  0.746779261

```

```

## [369,] 1.505734225 1.478499968
## [370,] -0.732266364 1.433724536
## [371,] 3.162587440 2.523386800
## [372,] -1.001661904 -0.358038669
## [373,] 0.647950731 0.048399790
## [374,] 0.763041716 2.166558142
## [375,] 0.199898001 2.183677489
## [376,] -0.985866058 1.464854949
## [377,] -0.762037774 0.535562884
## [378,] -0.296513497 2.046752758
## [379,] -0.135602538 1.610821371
## [380,] -0.984491575 -0.616366487
## [381,] 0.291403483 1.268440721
## [382,] 1.150741392 0.678747180
## [383,] 0.743918583 1.058716414
## [384,] -1.024153414 0.696810294
## [385,] 0.641701443 1.682985586
## [386,] 1.465198094 2.422555425
## [387,] 1.414177835 0.876046795
## [388,] -0.181191848 -0.285705079
## [389,] 0.505022963 0.240416949
## [390,] 0.602086592 1.497382462
## [391,] 0.434895861 -0.055611769
## [392,] -1.649597234 -0.587555843
## [393,] 1.823991879 1.207575467
## [394,] -1.132613913 1.195916546
## [395,] -0.227013440 0.733577910
## [396,] 2.152917938 1.951914326
## [397,] -0.290952141 0.450093791
## [398,] 1.408651598 1.556198399
## [399,] -0.777357379 0.093790581
## [400,] 0.071190913 0.814723632
## [401,] 0.943355705 0.429241532
## [402,] -0.116424882 -0.407738761
## [403,] -0.199504965 0.600088720
## [404,] -0.344503472 1.794548437
## [405,] 0.538531243 2.123148251
## [406,] -0.617131500 -0.741698974
## [407,] -0.215857507 1.110065395
## [408,] 1.179351368 1.630783015
## [409,] -0.117279246 0.684034406
## [410,] 0.566659833 1.327021324
## [411,] -0.361442436 2.714007328
## [412,] 1.324596333 0.639307522
## [413,] 1.015277816 1.616197675
## [414,] 1.283719826 0.003742538
## [415,] -0.153172331 0.936663592
## [416,] -2.337338374 -0.740729903
## [417,] -0.303320404 1.642341837
## [418,] 1.835210104 0.410064852
## [419,] 0.757284249 1.670053746
## [420,] -0.777570134 1.315799817
## [421,] 2.248102657 2.334086794
## [422,] 0.854126090 1.141975558
## [423,] 0.627661353 1.091124131
## [424,] 1.419305976 2.284324727

```



```

## [425,] -0.454789937 0.710369938
## [426,] -0.548915130 1.268905753
## [427,] 2.623998547 3.028510957
## [428,] 0.411499136 1.107777182
## [429,] 0.670712813 3.658710690
## [430,] -0.961332762 -0.030233834
## [431,] 0.240631055 0.929384164
## [432,] 1.498846277 0.656603151
## [433,] 0.758412978 1.261269026
## [434,] -0.260243471 0.018430376
## [435,] 0.522201936 1.011156106
## [436,] -0.145364836 -0.220782477
## [437,] 1.460627011 1.273811359
## [438,] 1.799829601 1.579055969
## [439,] -1.319643611 -1.034148126
## [440,] 0.886075606 2.754208254
## [441,] -1.099319375 0.449767252
## [442,] 1.073829299 1.213854181
## [443,] 1.914450676 0.728534221
## [444,] 0.749182045 0.404598617
## [445,] 0.170195192 0.212434658
## [446,] 0.591353202 1.194991206
## [447,] 1.211331780 1.048724505
## [448,] 0.261622336 2.768303027
## [449,] -1.141242198 -0.913509732
## [450,] -0.310173781 -1.024118251
## [451,] 1.809395644 2.168133886
## [452,] 1.909095961 1.403718379
## [453,] 0.798893434 1.455049141
## [454,] 1.002545862 1.736179476
## [455,] 1.116502973 1.972090584
## [456,] -0.993497611 -2.115355871
## [457,] 1.351172525 2.071883816
## [458,] -0.075236619 0.735213461
## [459,] 1.410274258 0.489165810
## [460,] -0.233296974 1.222068666
## [461,] 0.863058362 2.147229241
## [462,] -0.951595674 1.848023687
## [463,] 0.870685940 1.527388927
## [464,] 0.138290120 0.685023647
## [465,] -0.494674401 0.101911433
## [466,] 1.513016725 2.082782956
## [467,] 2.065874619 0.717401844
## [468,] 2.828818431 1.657691373
## [469,] 0.595194444 1.017655477
## [470,] 1.764814647 1.683747267
## [471,] 2.183394821 2.738131346
## [472,] 1.311833524 -0.299374849
## [473,] 0.103639856 -0.895058399
## [474,] 0.843457182 0.241973743
## [475,] -0.178394262 1.307663959
## [476,] 1.180604174 0.582114167
## [477,] 2.382656586 2.083142633
## [478,] 0.177614194 0.757488013
## [479,] 0.984404689 1.161658327
## [480,] 0.229611226 0.876028405

```

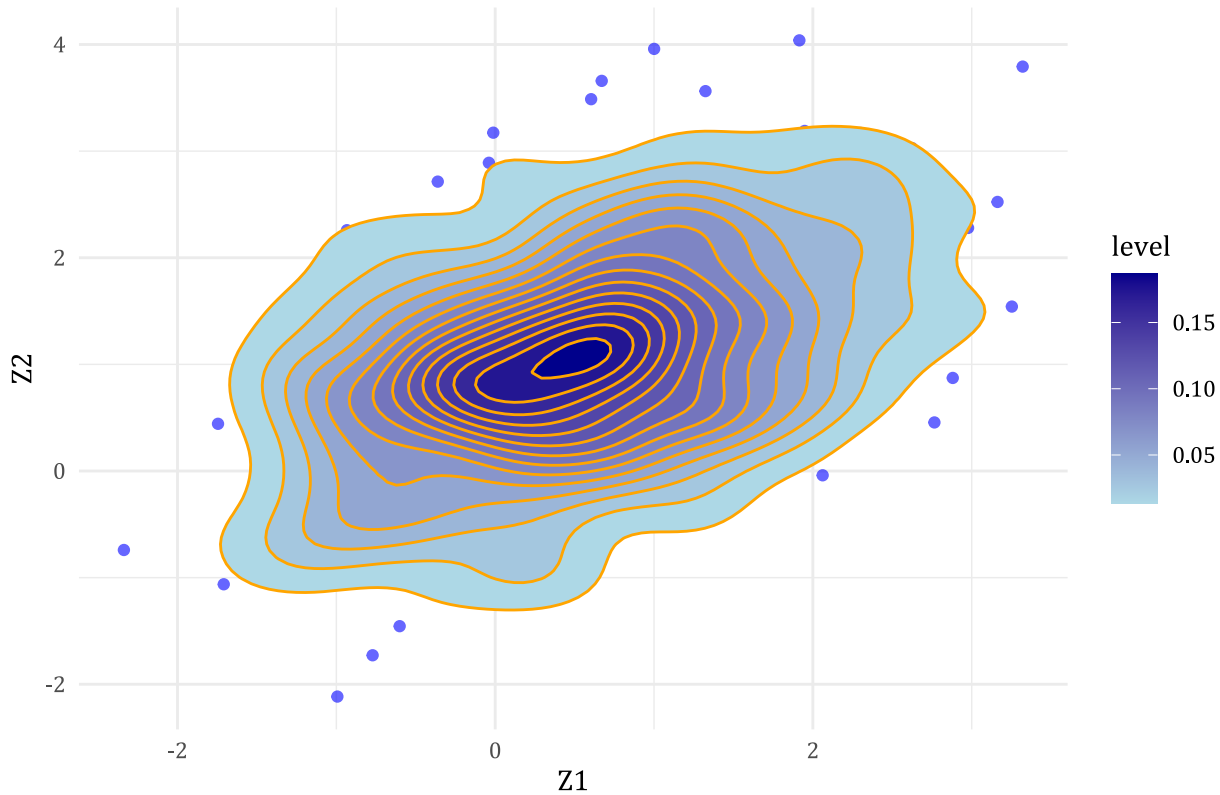
```
## [481,] -0.634820158 2.170241824
## [482,] 1.334283750 0.709680079
## [483,] 1.803002488 1.997532848
## [484,] 0.863444522 0.846685072
## [485,] 0.361141845 2.373537271
## [486,] 1.559255033 1.289770110
## [487,] 1.333260343 1.751173439
## [488,] -0.391007081 0.896126533
## [489,] 2.438215728 1.879625192
## [490,] -0.657506122 1.497669694
## [491,] 0.070507016 1.246269431
## [492,] 2.253046505 1.678839895
## [493,] 1.393420633 2.347998442
## [494,] -0.337307080 -0.050614223
## [495,] -0.412716052 0.400512720
## [496,] -0.631143367 -0.221118564
## [497,] 0.122226270 1.692742307
## [498,] 1.318306572 0.467206186
## [499,] 1.076640918 1.054024221
## [500,] 1.102777946 1.353587269
```

- b. Với dữ liệu vừa sinh ra ta biểu diễn dưới dạng đồ thị tán xạ kết hợp cùng với các đường mật mức. Để vẽ đồ thị ta sẽ sử dụng gói lệnh **ggplot2**.

```
library(ggplot2)
library(showtext)

## Loading required package: sysfonts
## Loading required package: showtextdb
font_add("LM", "C:/Windows/Fonts/cambria.ttc")
showtext_auto()
sample_df <- data.frame(Z1 = samples[, 1], Z2 = samples[, 2])
ggplot(sample_df, aes(x = Z1, y = Z2)) +
  geom_point(color = "blue", alpha = 0.6) +
  stat_density_2d(
    aes(fill = after_stat(level)),
    geom = "polygon",
    color = "orange",
    bins = 15
  ) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(
    title = "Đồ thị tán xạ cùng đường mật mức",
    x = "Z1",
    y = "Z2"
  ) +
  theme_minimal(base_family = "LM")
```

Đồ thị tán xạ cùng đường mặt mức



Để sinh dữ liệu mô phỏng phân bố chuẩn nhiều chiều ta dùng thư viện **mvtnorm**

```
library(mvtnorm)
set.seed(1)
x <- rmvnorm(50, mean = c(0, 0), sigma = matrix(c(1, .8, .8, 1), nrow = 2))
# Sinh ngẫu nhiên 50 quan sát có phân phối chuẩn nhiều chiều N(mean, sigma)
head(x)
```

```
##           [,1]      [,2]
## [1,] -0.47818953 -0.11590308
## [2,] -0.03397769  1.05315805
## [3,] -0.07220391 -0.58648888
## [4,]  0.76615864  0.87836259
## [5,]  0.37842066 -0.01565043
## [6,]  1.52652138  1.02477548
```

```
colMeans(x) # Trung bình mẫu: là 1 vector 2 giá trị
```

```
## [1] 0.1848335 0.1073419
```

```
cov(x) # Ma trận hiệp phương sai mẫu: là ma trận vuông
```

```
##           [,1]      [,2]
## [1,] 0.8373082 0.7040840
## [2,] 0.7040840 0.8705075
```

```
cor(x) # Ma trận tương quan mẫu: là ma trận vuông
```

```
##           [,1]      [,2]
## [1,] 1.0000000 0.8246992
## [2,] 0.8246992 1.0000000
```

Ngoài ra có thể sử dụng gói lệnh **mnormt**

```
library(mnormt)
```